Programming Fundamentals

Arrays

Arab Academy
for Science, Technology and Maritime Transport

Lecture Outline

- What is an array?
- Defining Arrays
- Arrays input/output
- Examples

What is an Array?

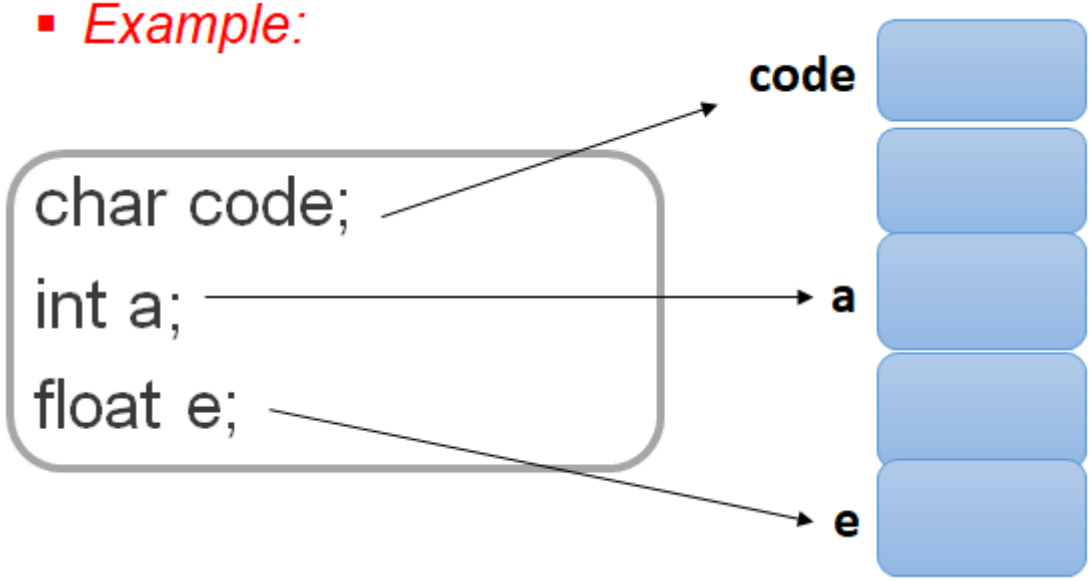
Array is a group of data that holds fixed number of value all of them are the same type .

Remember: Define a Variable

Syntax

Data_Type Variable_Name;

- *Example:*



2. Defining an Array

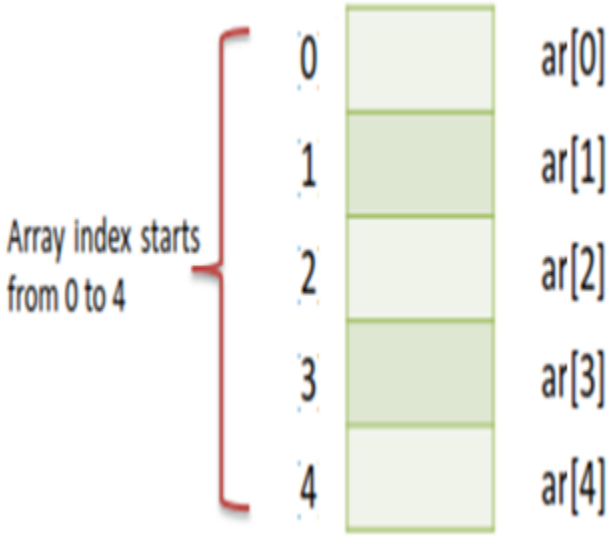
Syntax

Array_Type Array_Name[Length];

Example

```
int ar[5];
```

This line creates an array of *int* of size *5*.



2. Defining an Array(initialization)

Example

```
int ar[5]= {1,2,3,4,5};
```

1	ar[0]
2	ar[1]
3	ar[2]
4	ar[3]
5	ar[4]

```
int ar[5]= {1,2};
```

1	ar[0]
2	ar[1]
0	ar[2]
0	ar[3]
0	ar[4]

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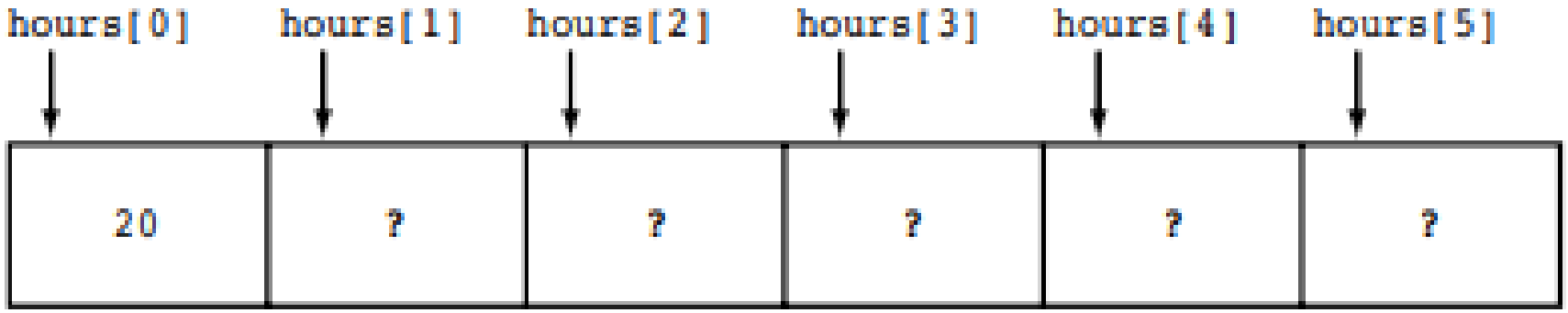
Defining an Array(initialization)

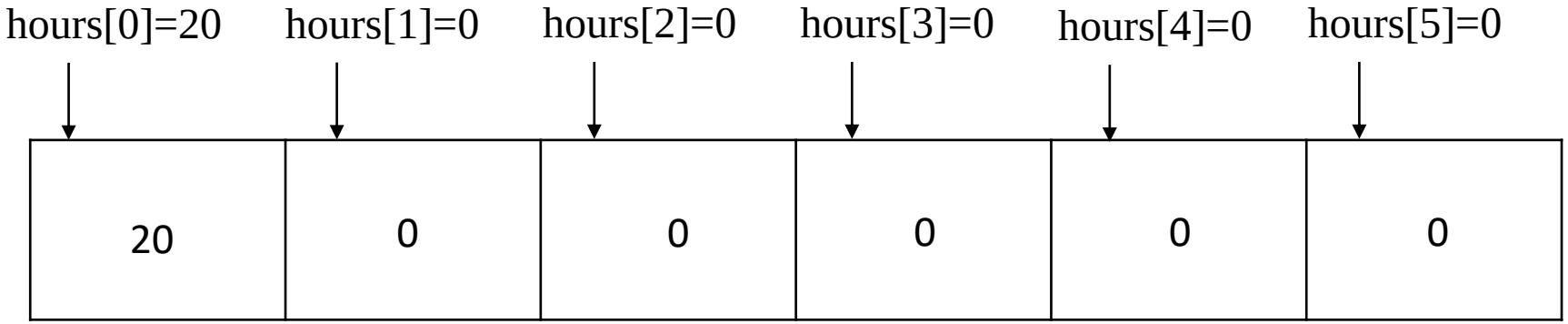
Example(1):

```
hours[0] = 20;
```



Example(2):

hours[5]={20};

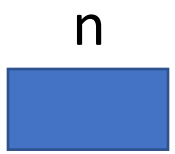


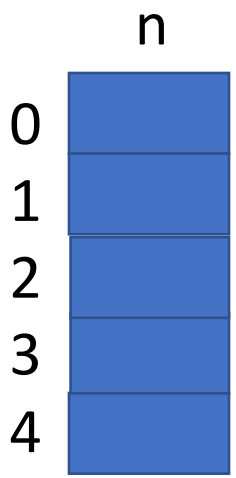
Remember: Input a Variable

```
int n;
cin>>n;
```

Could we input an array in the same way?!

```
int n[5];
cin>>n;
cin>>n[0];
cin>>n[1];
cin>>n[2];
cin>>n[3];
cin>>n[4];
```





Arrays Input

In order to input the values from the user and store it in an array, the for loop is used to save the values one by one

```
int i, n[10];
for ( i = 0; i < 10; i++ )
        { cin>> n[ i ] ; }
```

```
cin>>n[0];
cin>>n[1];
cin>>n[2];
cin>>n[3];
.....
.....
cin>>n[9];
```

Arrays Output

In order to display the values of an array, the for loop is used.

```
for ( i = 0; i < 10; i++ )
        { cout<<n[ i ]; }
```

Example: find the sum of given 4 numbers

```
#include <iostream>
using namespace std;

int main() {
  // initialise array
  int arr[4] = {2, 4, 6, 8};

  // initialise sum to zero
  int sum = 0;

  for(int i = 0; i < 4; i++)
  {
    sum = sum + arr[i];
  }

  cout << "The sum = " << sum;
}
```

arr[0]=2
arr[1]=4
arr[2]=6
arr[3]=8

Trace Table

i	arr[i]	Sum
0	2	0
1	4	2
2	6	6
3	8	12
		20

Example: find the sum of any 4 numbers

```
#include <iostream>
using namespace std;

int main() {
  // declare array
  int arr[4];
// read the array
  for(int i = 0; i < 4; i++)
  {
    cin>>arr[i];
  }

  // initialise sum to zero
  int sum = 0;

  for(int i = 0; i < 4; i++)
  {
    sum = sum + arr[i];
  }

  cout << "The sum = " << sum;
}
```

i	arr[i]
0	2
1	8
2	4
3	6

Size	i	arr[i]	Sum
4	0	2	0
	1	8	2
	2	4	10
	3	6	14
			20

THANK YOU